

MULTIPLE CHOICE

- 1) Assume \$t0 holds the value 0x00101000 (=40). What is the value of \$t2 after the following instructions?

```
slt $t2, $t0, $0
bne $t2, $0, ELSE
j DONE
ELSE: addi $t2, $t2, 2
DONE:
```

- A) 2
☒ B) 0
 C) 40
 D) 1
- 2) Which instruction must be wrong for calculating $a = b + c + d - e$?
- A) add t0, t0, b
 B) add t0, t0, c
☒ C) sub a, e, t0
 D) sub a, t0, e

- 3) What decimal number does the hexadecimal number A0 represent if it is an unsigned integer?

A) 140
 B) 100
 C) 170
☒ D) 160

- 4) What is the decimal value for the following single-precision floating-point number?



- A) -1
☒ B) 1
 C) 0
 D) 2
- 5) In floating point standard, we use a bias to adjust exponent. If the exponent is 4-bit, then the bias is equal to
- A) 16
☒ B) 7
 C) -16
 D) -7
- 6) For hexadecimal numbers, what is 5ED4 - 07A4?

A) 5620
 B) 5720
 C) 5630
☒ D) 5730

7) Assume \$1=100 and \$2=-50 are signed 8-bit decimal integers. When you do \$1-\$2, what do you get?

- ☒ A) Overflow
- B) Legitimate result 150
- C) Underflow
- D) None of above

8) In floating point standard, we use a bias to adjust exponent. For single-precision (8-bit exponent), the bias is equal to

- ☒ A) 127
- B) -127
- C) -128
- D) 128

9) Let binary numbers A = 00001011, B = 11100101, what is A+B in decimal?

- A) -15
- B) 0
- ☒ C) -16
- D) -14

10) Assume the following register contents:

\$t0 = 0xFFFFFFFF, \$t1 = 0x12345678

What is the value of \$t2 for the following sequence of instructions?

sll \$t2, \$t0, 4

or \$t2, \$t2, \$t1

- A) 0FFFFFFF
- B) FFFFFFFF
- ☒ C) FFFFFFF8
- D) 0FFFFFF8

11) Assume A=base of array A, B=base of array B. What does the following code do?

sub \$t0, \$i, \$j

sll \$t0, \$t0, 2

add \$t0, \$B, \$t0

lw \$t1, 0(\$t0)

sw \$t1, 32(A)

- ☒ A) A[8] = B[i-j];
- B) B[8] = A[4*(i-j)];
- C) B[8] = A[i-j];
- D) B[32] = A[i-j];

- 12) Assume that registers \$s0 and \$s1 hold signed integers 0x80000000 and 0xD0000000 (in hex), respectively. What is the value of \$t0 for the following assembly code?

```
sub $t0, $s0, $s1
```

- ☒ A) Overflow
 - ☐ B) B0000000
 - ☐ C) -B0000000
 - ☐ D) None of above
- 13) In ASCII-code, 'A'=65. Assume memory at the indicated addresses have the given decimal values:
- 2000: 'A'
2001: 'B'
2002: 'C'
2003: 'D'

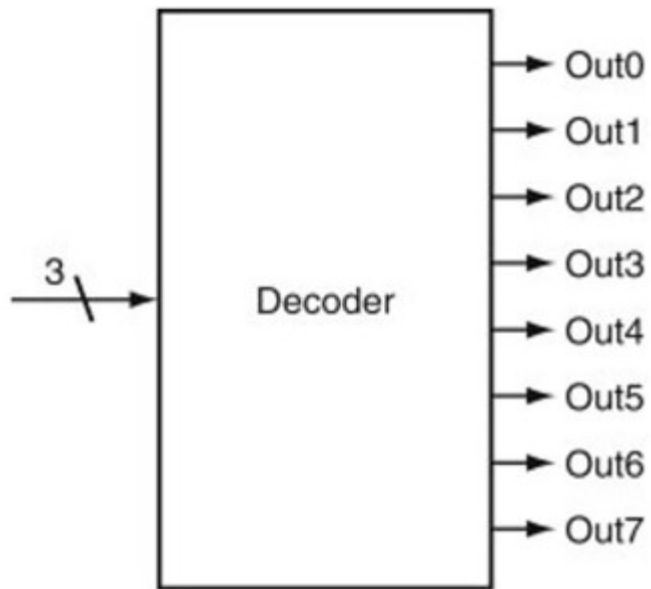
If \$t1 holds 2000, what is \$t2 (in decimal) after lw \$t2, 0(\$t1)

- ☐ A) 65
 - ☐ B) 65666768
 - ☒ C) A very large number
 - ☐ D) 65 + 66 + 67 + 68
- 14) Suppose \$t0 is a pointer to an array of integer. To increment the pointer to next integer, which instruction is not correct?
- ☐ A) add \$t0, \$t0, \$t1 # \$t1 contains 4
 - ☐ B) addi \$t0, \$t0, 4
 - ☒ C) addi \$t0, \$t0, 1
 - ☐ D) sll \$t0, \$t0, 2

- 15) Assume \$t0 holds the value 0x00101000 (=40). What is the value of \$t2 after the following instructions?

```
slt $t2, $0, $t0
bne $t2, $0, ELSE
j DONE
ELSE: addi $t2, $t2, 2
DONE:
```

- ☒ A) 3
- ☐ B) 1
- ☐ C) 0
- ☐ D) 2

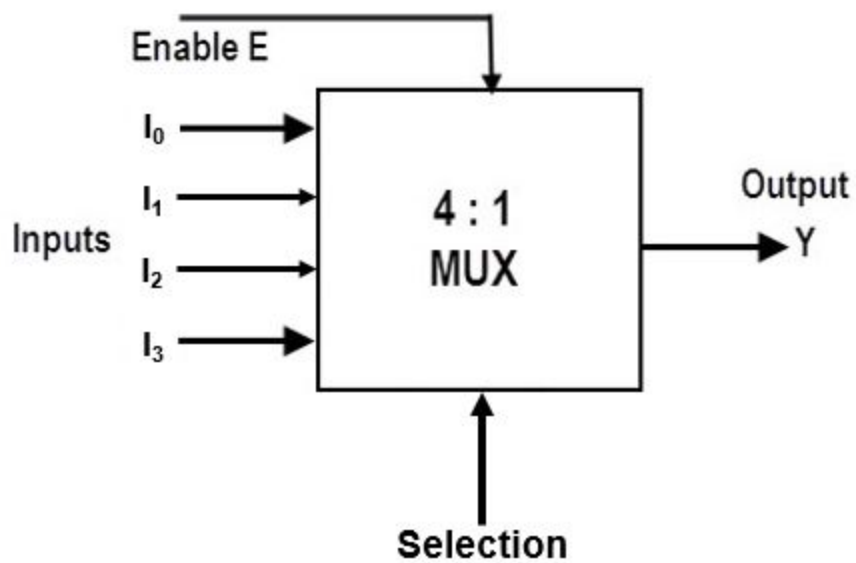


a. A 3-bit decoder

16)

Let the 3-bit decoder's input = 000, the output bits will have

- A) No "1" and eight "0"
- ☒ B) One "1" and seven "0"
- C) Eight "1" and no "0"
- D) Seven "1" and one "0"



17)

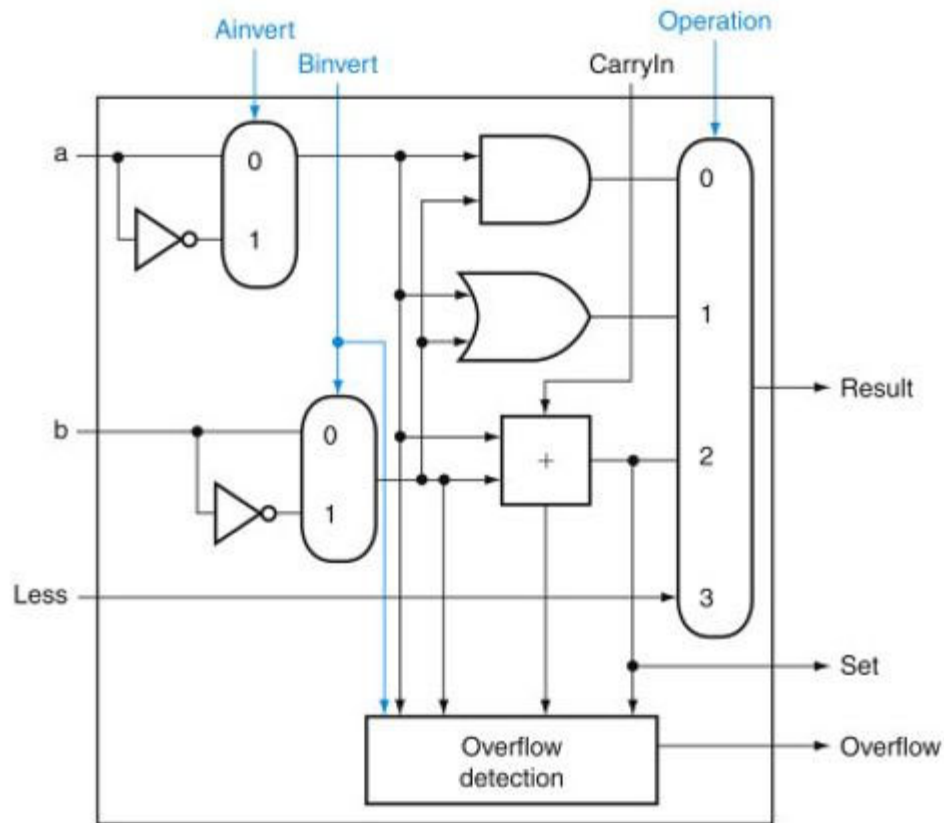
In the above multiplexer, the Selection has ____ lines

- A) 4
- B) 3
- ☒ C) 2
- D) 1

18) Consider instruction:

lw \$t1, -50000(\$t0)

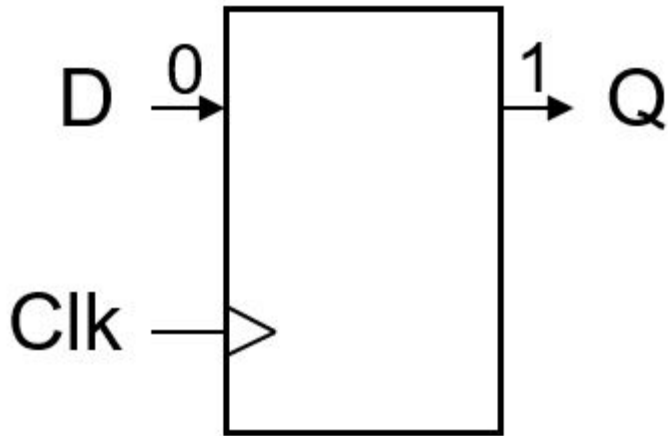
- A) It is illegal because \$t0 always contains zero
- ✓ B) It is illegal because the offset is too big
- C) It is illegal because the offset is negative
- D) It is a legal instruction



19)

Given the above 1-bit digital circuit, what's the control code for operation set-less-than?

- A) 0011
- B) 0010
- C) 0111
- D) 0110



20)

Given the above 1-bit register. What's D's and Q's values after the next Clock cycle begins?

- A) D=0; Q=1
- B) D=0; Q=0
- C) D=1; Q=1
- D) None above

Answer Key

CSIT545MIDTERM_SP2025-SAMPLEQUESTIONS(20)

- 1) B
- 2) C
- 3) D
- 4) B
- 5) B
- 6) D
- 7) A
- 8) A
- 9) C
- 10) C
- 11) A
- 12) B
- 13) C
- 14) C
- 15) A
- 16) B
- 17) C
- 18) B
- 19) C
- 20) B

Midterm Open-ended Questions with Solutions

(Fall 2022 Midterm Questions)

1. The following MIPS instruction sequence could be used to implement an operation that involves a register-pair (\$s0-\$s1).

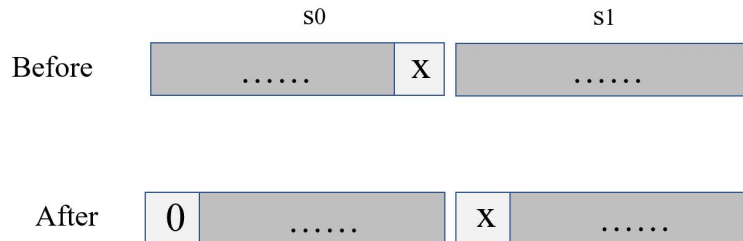
```
sll $t0, $s0, 31 (Keep $s0's rightmost bit in $t0's leftmost bit)
srl $s0, $s0, 1 (Shift-right 1 bit of $s0 (rightmost bit shifted out) )
srl $s1, $s1, 1 (Shift-right 1 bit of $s1)
or  $s1, $s1, $t0 ($s0's original rightmost bit or'ed to the beginning of $s1)
```

Register \$t0 is just being used as a temporary holder.

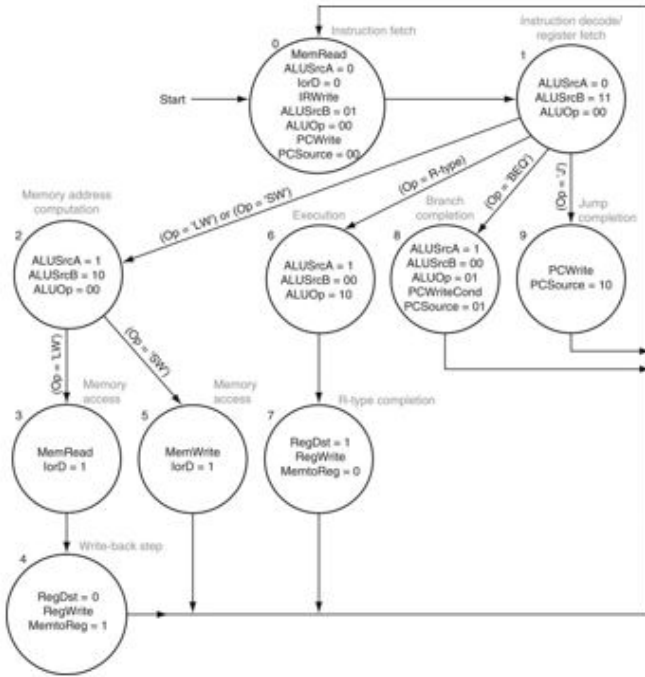
What operation does the instruction sequence accomplish? Draw diagrams to show operation if you feel necessary.

Answer:

It accomplishes a shift-right 1 bit, or divide by 2, for the 64-bit word \$s0-\$s1.



2. Given the Finite State Machine representing our smallMIPS processor, and the NextState signals, write the Boolean expression for NS_3 .



	NS_3	NS_2	NS_1	NS_0
NextState0	0	0	0	0
NextState1	0	0	0	1
NextState2	0	0	1	0
NextState3	0	0	1	1
NextState4	0	1	0	0
NextState5	0	1	0	1
NextState6	0	1	1	0
NextState7	0	1	1	1
NextState8	1	0	0	0
NextState9	1	0	0	1

Answer:

$$\begin{aligned}
 NS_3 &= \text{NextState8} + \text{NextState9} \\
 &= \text{State1} \cdot [\text{Op} = \text{'BEQ'}] + \text{State1} \cdot [\text{Op} = \text{'J'}] \\
 &= \overline{S_3} \overline{S_2} \overline{S_1} S_0 \cdot ([\text{Op} = \text{'BEQ'}] + [\text{Op} = \text{'J'}])
 \end{aligned}$$